

Wentao DONG

Data Security, Network Security, Privacy-Enhancing Technologies

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PERSONAL STATEMENT

I am currently a postdoctoral researcher at the the Laboratory for AI-Powered Financial Technologies (AIFT) in Hong Kong, mentored by Prof. [Xiaohua JIA](#) (ACM/IEEE Fellow). I received my Ph.D. in computer science from City University of Hong Kong, advised by Prof. [Cong WANG](#) (IEEE Fellow), and my B.Eng. degree in computer science from Shanghai Jiao Tong University. My research focuses on building efficient privacy-enhancing technologies (PETs), spanning horizontally across multi-party computation (MPC), trusted execution environment (TEE), differential privacy (DP), zero-knowledge proof (ZKP), and federated learning (FL). I also work to extend this line of research vertically—up to the application layer to secure AI and FinTech systems, and down to the network and compute layers for scalable cryptographic computation. Beyond academia, I am also committed to the industrial collaboration of cutting-edge privacy protection research. I have secured fundings from multiple innovation programs, pushing privacy technology from theoretical research to practical application.

EDUCATION BACKGROUND

- **Shanghai Jiao Tong University—B.Eng. in Computer Science and Technology** 2014—2018
Thesis Title: Post-Quantum Hash-Based Digital Signature: Implementation and Application. Shanghai
- **City University of Hong Kong—Ph.D. in Computer Science** 2021—2025
Thesis Title: Building Privacy-Preserving Systems from Communication-Efficient Secure Multi-Party Protocols. Hong Kong SAR

WORK EXPERIENCE

- **Intel Asia-Pacific Research & Development Ltd.** 2018—2019
Position: R&D Engineer | Non-Volatile Memory Solutions Group (NSG) and Data Center Group (DCG) Shanghai
- **Sangfor Technologies Inc.** 2019—2020
Position: Software Engineer | Security Business Group (Security BG) Shenzhen
- **City University of Hong Kong & Monash University (Remote)** 2020—2021
Position: Research Assistant | Department of Computer Science & Faculty of Information Technology Hong Kong SAR & Melbourne
- **Laboratory for AI-Powered Financial Technologies (AIFT)** 2025—Present
Position: Postdoctoral Researcher Hong Kong SAR

SELECTED PUBLICATIONS

J=JOURNAL, C=CONFERENCE, P=PREPRINT

- [J.1] **Wentao Dong**, Lei Xu, Leqian Zheng, Huayi Duan, Cong Wang, and Qian Wang, “Do Not Skip Over the Offline: Verifiable Silent Preprocessing from Small Security Hardware,” *IEEE Transactions on Information Forensics and Security (TIFS)*, Vol. 20, pp. 4860-4873, 2025.
- [J.2] Peipei Jiang, Yihao Wu, Lei Xu, **Wentao Dong**, Peiyuan Chen, Yulong Ming, Cong Wang, Xiaohua Jia, and Qian Wang, “Anonymous Messaging Made More Flexible with PingPong,” *IEEE/ACM Transactions on Networking (TON)*, Vol. 34, pp. early access, 2026.
- [C.1] **Wentao Dong**, Yang Cao, Cong Wang, and Wei-Bin Lee, “DOPPIO: Communication-Efficient and Secure Multi-Party Shuffle Differential Privacy,” *International Conference on Very Large Data Bases (VLDB)*. 2026.
- [C.2] **Wentao Dong**, Peipei Jiang, Huayi Duan, Cong Wang, Lingchen Zhao, and Qian Wang, “Ring of Gyges: Accountable Anonymous Broadcast via Secret-Shared Shuffle,” *Network and Distributed System Security (NDSS) Symposium*. 2025.
- [C.3] **Wentao Dong** and Cong Wang, “Poster: Towards Lightweight TEE-Assisted MPC,” *ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2023.
- [C.4] Leqian Zheng, Zheng Zhang, **Wentao Dong**, Yao Zhang, Ye Wu, and Cong Wang, “H₂O₂RAM: A High-Performance Hierarchical Doubly Oblivious RAM,” *USENIX Security Symposium (SECURITY)*. 2025.
- [C.5] Yichen Zang, Chengjun Cai, **Wentao Dong**, Lei Xu, and Cong Wang, “HiddenTor: Toward a User-Centric and Private Query System for Tor BridgeDB,” *IEEE International Conference on Distributed Computing Systems (ICDCS)*. 2024.

- [C.6] Yichen Zang, Chengjun Cai, **Wentao Dong**, and Cong Wang, “VizardFL: Enabling Private Participation in Federated Learning Systems,” *International Web Information Systems Engineering Conference (WISE)*. 2024.
- [P.1] Zikai Mao, Lingchen Zhao, Lei Xu, **Wentao Dong**, Shenyi Zhang, Cong Wang, and Qian Wang, “Amulet: Fast TEE-Shielded Inference for On-Device Model Protection,” arXiv preprint arXiv:2512.07495 (2025).

PRESENTATIONS AND TALKS

O=ORAL, P=POSTER

- [O.1 & P.1] H₂O₂RAM: A High-Performance Hierarchical Doubly Oblivious RAM. Presented in *USENIX Security Symposium (Security)*. 2025 Aug. 15. Seattle, WA, USA.
- [O.2] *Ring of Gyges*: Accountable Anonymous Broadcast via Secret-Shared Shuffle. Presented in *Network and Distributed System Security (NDSS) Symposium*. 2025 Feb. 27. San Diego, CA, USA.
- [O.3] *Architecting the Foundation of the Trusted Data Spaces*: A Full-Stack Perspective on Secure Computation Protocols. Presented in *6th SJTU Paris Elite Institute of Technology Young Scholar Forum*. 2025 Nov. 20. Shanghai, China.
- [P.2] Poster: Towards Lightweight TEE-Assisted MPC. Presented in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 2023 Nov. 26. Copenhagen, Denmark.

GRANTS AND PROPOSALS (PARTICIPANT)

- **Collaborative Research Fund (CRF), University Grants Committee (UGC)** PI: Prof. Cong WANG
Project Title: Enabling Metadata-private and Accountable Networks at Scale CityUHK
- **Research Impact Fund (RIF), University Grants Committee (UGC)** PI: Prof. Xiaohua JIA
Project Title: Secure and Incentivized Data Sharing with Enhanced Ownership for Decentralized Networks CityUHK
- **General Research Fund (GRF), University Grants Committee (UGC)** PI: Prof. Cong WANG
Project Title: Enabling Multi-Party Shuffle Differential Privacy: System Modeling and Security Hardening CityUHK
- **General Research Fund (GRF), University Grants Committee (UGC)** PI: Prof. Cong WANG
Project Title: Towards Bridging the Last-mile Gap of Multi-party Collaborative Statistics CityUHK
- **“Strategic S&T Innovation Cooperation” Key Specialized Program** PI: Prof. Chao SHE & Prof. Cong WANG
Project Title: AI Algorithm Security Evaluation and Privacy Enhancement for Financial Applications XJTU & CityUHK

TECHNOLOGY TRANSFER PROJECTS (PARTICIPANT)

- **Data Security: X-Crypt@HK** Person-in-Charge
Project Title: Integrated Privacy-Enhancing Computation and Data Protection Suite Hong Kong SAR
- **Network Security: Synapiron@HK&GZ** Core Member
Project Title: Building Trusted Data Spaces for Secure and Compliant Data Flow Hong Kong SAR & Guangzhou

TEACHING SERVICES

- **Teaching Assistant** 2021—2023
CS3504 IT Professional Placement & CS3402 Database Systems CityUHK
- **Teaching Assistant/Course Developer/Tutor** 2023—2025
CS2311 Computer Programming CityUHK

COMMUNITY SERVICES

- **Academic Reviewer**
IEEE IoT-J ('23), TSC ('24-'25), TDSC ('25), MSN ('25), VLDB('26)
- **External Reviewer**
IEEE S&P ('24), TDSC ('23), INFOCOM ('21), ICDCS ('22), ICNP ('25), USENIX Security ('22-'25), ACM MM ('25), AsiaCCS ('21), etc.
- **Student Helper**
IEEE ICDCS ('23)